

Hard equation categories - GPT-5 trace spec

Generated from known rules and solver-verified; the 'answer' is ground truth. Your trace's boxed answer must match it.

Solver accuracy on competition problems

cryptarithm_deduce	54 / 659	8.2%
cryptarithm_guess	11 / 164	6.7%
equation_numeric_deduce	540 / 596	90.6%
equation_numeric_guess	?	low

equation_numeric_deduce

Visible two-digit operands joined by a single SYMBOL operator, e.g. ``13{45 = 1345``. Each operator symbol stands for ONE arithmetic/string operation that is consistent across all examples sharing that symbol. 'deduce' uses COMMON operations: concatenation, reverse concatenation, addition, absolute difference, subtraction, reverse subtraction, multiplication. Infer the operation from the examples, then apply it to the query.

In Alice's Wonderland, a secret set of transformation rules is applied to equations. Below are a few examples:

`13{45 = 1345`

`41{38 = 4138`

`27{23 = 2723`

`96{79 = 9679`

Now, determine the result for: `21{85`

ANSWER: 2185

In Alice's Wonderland, a secret set of transformation rules is applied to equations. Below are a few examples:

`13|21 = 8`

`37|39 = 2`

`74|87 = 13`

`13|81 = 68`

Now, determine the result for: `35|93`

ANSWER: 58

equation_numeric_guess

Same visible-digit format, but the operation is RARE/digit-wise: digit add mod10, digit sub mod10, digit multiply, cross multiply, determinant ($d1*d4 - d2*d3$), modulo, integer division. Harder to spot; test digit-position operations explicitly. Example ``48/69 = 07`` is digit add mod10: $(4+6)\%10=0$, $(8+9)\%10=7$.

In Alice's Wonderland, a secret set of transformation rules is applied to equations. Below are a few examples:

`48/69 = 07`

`96/79 = 65`

`64/78 = 32`

`58/39 = 87`

Now, determine the result for: `41/68`

ANSWER: 09

In Alice's Wonderland, a secret set of transformation rules is applied to equations. Below are a few examples:

`86{14 = 72`

`67{86 = 81`

`96{16 = 80`

`41{15 = 36`

Now, determine the result for: `61{66`

ANSWER: 05

cryptarithm_deduce

Like the equation format, but every DIGIT is replaced by a fixed unique SYMBOL (a 10-symbol substitution). The rule here is FORWARD concatenation of the two encoded operands. You do NOT need to decode the digits ? operate on the symbols directly: output = left-operand-symbols followed by right-operand-symbols.

In Alice's Wonderland, a secret set of transformation rules is applied to equations. Below are a few examples:

``=#** = `=#*`
`|+#%* = |+%*`
`({#;` = ({;``
`*`#=# = *`=#`

Now, determine the result for: ``;#;``

ANSWER: ``;;``

In Alice's Wonderland, a secret set of transformation rules is applied to equations. Below are a few examples:

`}#]#( = }##(`
`(*),` = (*,``
`+[+] = +(+;`
`=`]~; = =`~;`

Now, determine the result for: `++]++#`

ANSWER: `+++#`

cryptarithm_guess

Symbol-encoded operands with REVERSE concatenation: output = right-operand-symbols followed by left-operand-symbols. Operate on the symbols directly.

In Alice's Wonderland, a secret set of transformation rules is applied to equations. Below are a few examples:

`;@|^*^ = *^;@`
`^*|. = .;^*`
`*_|@. = @.*_`
`$$|;] = ;]$$`

Now, determine the result for: `$;|$@`

ANSWER: `$@$;`

In Alice's Wonderland, a secret set of transformation rules is applied to equations. Below are a few examples:

`&{])[ = )[&[`
`[#{][, = [, [#{`
`.)}&; = &;.)`
`[&}.; = .;[&`

Now, determine the result for: `)&}_)`

ANSWER: `_))&`

Answer contract

Output format (mandatory): put all reasoning inside one `<think> ... </think>` block, then immediately the final answer ONCE as `\boxed{...}` with nothing after it. The boxed content is the answer only, in the exact form the prompt expects (a number, a digit string with leading zeros preserved, or the symbol string) ? no spaces, no operator, no extra words. Answers are checked with the official metric: numeric within 1e-2 relative tolerance, otherwise exact case-insensitive string match.